Literal automorphism and correlation groups for the $\mathrm{PSL}_3(4)$ geometries

L100 structural certificate

This note identifies the groups attached to the 29 geometries in the fixed 22-page Halle preprint of Gottschalk and Leemans [1]. It acquires the complete literal coset models, their proved source-label identifications, and their complete palette automorphism groups. It does not repeat the ambient subgroup or geometry census or the GRAPE/nauty computation. The completeness of the latter groups remains an explicit acquired computational premise. The journal edition has not been independently collated.

Notation and acquired actions. Put $G = \mathrm{PSL}_3(4)$, in the retained 21-point action of order 20160. Its two actual generators, all its elements, and their actions on every coset model are acquired literally. Each palette graph has the objects and one additional vertex per type; objects and palette vertices have separate colours. Its full colour-preserving automorphisms are precisely $\mathrm{Cor}(\Gamma)$; the subgroup fixing every palette vertex is $\mathrm{Aut}(\Gamma)$. These statements use the accepted palette lemma, rather than the uncoloured incidence graph. Both groups contain the faithful embedded G.

On coordinates over $\mathbb{F}_4$, let δ be conjugation by $\mathrm{diag}(\omega, 1, 1)$, σ the field map $a \mapsto a^2$, and τ the inverse-transpose map induced by ordinary projective polarity. The explicit coordinate-to-retained-point bijection is acquired. In particular τ uses the bilinear dot product, not a Hermitian polarity. The twelve maps $\delta^i\sigma^j\tau^k$ form an actual subgroup

$$O = \langle \delta, \sigma, \tau \rangle \cong S_3 \times C_2, \qquad 0 \le i < 3, \quad 0 \le j, k < 2.$$

Indeed, both involutions invert δ and commute with each other. The certificate checks the complete multiplication table on the two generators of G. It checks that all 241920 maps obtained by following one of these twelve maps by conjugation by an element of G have distinct generator-image pairs. Thus O intersects the inner group trivially. The separately acquired complete computation $|\mathrm{Aut}(G)| = 241920$ identifies this explicit subgroup with all of $\mathrm{Aut}(G)$; the order is not inferred from the geometry table.

Define the following precise names:

$$\begin{array}{lll} \Sigma = G \rtimes \langle \sigma \rangle, & T = G \rtimes \langle \tau \rangle, & U = G \rtimes \langle \sigma\tau \rangle, \\ \Gamma = G \rtimes \langle \delta, \sigma \rangle, & V = G \rtimes \langle \sigma, \tau \rangle, & A = \mathrm{Aut}(G). \end{array} \tag{1}$$

Here Σ is the field-extension meaning of $P\Sigma L_3(4)$, explicitly documented by the primary Magma handbook [2], $\Gamma = P\Gamma L_3(4)$, and V supplies the explicit meaning of the source's $P\Sigma L_3(4) : 2$ in this table. Outer subgroups conjugate by O give the same names, with an actual conjugating element recorded.

Lemma 1. *For every saved typed or correlation group H, the certificate determines its conjugation image in $\mathrm{Aut}(G)$, its kernel, and an explicit splitting. It proves $H \cong G \rtimes E$ or $C_2 \times (G \rtimes E)$, for the recorded subgroup $E \le O$.*

Proof. Choose the smallest object type. Traverse the already known generators simultaneously on G and on this type. The 20160 restricted permutations are distinct, so this action is faithful. For each actual palette generator q and each generator g of G, look up the restriction of $q^{-1}gq$.

Then reconstruct the resulting element's action using its saved generator word and check equality on *every* object and palette vertex. Consequently every q normalizes the embedded G; restricted equality alone is never used as the conclusion. Its two generator images have a unique decomposition into an inner map and one of the twelve coordinate maps. The small quotient multiplication table determines E exactly.

The conjugation image contains all inner maps and projects onto E, so it is the entire preimage $G \rtimes E$, of order $20160|E|$. The acquired complete order of H therefore gives its kernel order, which is one or two in these 58 instances. This use of order follows the explicit image identification; it does not identify a group by order.

Choose short words in the actual generators lifting every element of E. Correct each lift by its exact inner factor, giving q_e whose conjugation on G is the specified coordinate map e. All products $q_e q_f q_{ef}^{-1}$ are checked literally. Discrepancies between the original generators and these normalized lifts are also checked. In the kernel-two cases their nonidentity values exhibit a single permutation z. The certificate verifies $z^2 = 1$, $z \neq 1$, and commutation with every generator of H and G. These relations detect the whole kernel: otherwise the normalized lifts together with G would already express every original generator with trivial kernel, contradicting the acquired order.

Write the multiplication discrepancies as $z^{c(e,f)}$. Solve the finite binary system

$$b(e) + b(f) + b(ef) = c(e, f) \pmod 2.$$

The certificate then checks all products of the actual corrected permutations $s_e = q_e z^{b(e)}$, including $s_1 = 1$. Thus these permutations form an actual homomorphic section. The subgroup generated by G and the s_e has the faithful conjugation image $G \rtimes E$ and has trivial intersection with $\langle z \rangle$. Together they generate H, proving the direct product assertion. In particular a central kernel of size two is never assumed to split without this section certificate. □

Theorem 2. *With notation (1), the literal models have the following groups. All five displayed direct factors C_2 are certified by the preceding lemma.*

Source	Aut(Γ)	Cor(Γ)	Preprint page
2.1	Γ	A	7
2.2	Σ	Σ	7
2.3	V	V	7
2.4	U	V	7
2.5	Σ	Σ	8
2.6	A	$C_2 \times A$	8
2.7	A	A	8
3.1	Σ	Σ	8
3.2	V	$C_2 \times V$	8
3.3	Σ	V	8
3.4	Σ	V	9
3.5	U	V	9
3.6	G	Σ	9
3.7	Γ	A	9
4.1	G	Σ	9
4.2	Σ	V	10
4.3	Σ	V	10
4.4	Σ	Σ	10
4.5	Σ	Σ	10
4.6	Σ	Σ	11
4.7	Σ	V	11
4.8	G	$C_2 \times \Sigma$	11
4.9	G	Σ	11

4.10	G	G	12
4.11	G	T	12
5.1	G	G	12
5.2	G	Σ	13
5.3	G	$C_2 \times \Sigma$	13
5.4	G	$C_2 \times G$	13

Proof. Apply the preceding finite certificate to the exact 29 acquired model/group pairs, with source labels supplied by the separately accepted literal source joins. Every generator is checked, not a proposed generating subset of the complete palette group. The supplementary data retain all outer/inner image pairs, lift words, normalization elements, section permutations and central involutions. The five nontrivial kernels occur only in the correlation groups of 2.6, 3.2, 4.8, 5.3 and 5.4. No typed group has a nontrivial kernel. □

Two precise name corrections. The preprint prints $P\Sigma L_3(4)$ for both Aut(3.5) on page 9 and Cor(4.11) on page 12, as it also does for groups of field type elsewhere. With the explicit field convention in (1), the first two names require replacement by U and T, respectively. This is not merely a relabelling of abstractly isomorphic groups. The acquired 20160-element list contains 315 involutions of G. For an involutory coordinate automorphism a, the involutions in its outer coset correspond exactly to

$$\{h \in G : h\, a(h) = 1\}.$$

Literal tests over this complete finite set give 120, 280 and 336 for $a = \sigma, \sigma\tau, \tau$, respectively. Thus Σ, U, T have 435, 595 and 651 involutions and are pairwise nonisomorphic. Even if one leaves the source's ATLAS-notation convention uninterpreted, its identical printed name cannot denote all three groups consistently. The unqualified $L_3(4) : 2$ at 2.4 is instead refined to U, not declared erroneous. All group orders remain unchanged, and these auxiliary name corrections do not refute the principal geometry classification.

Scope and attribution. The computations and proof concern only the complete typed and correlation group names. Abstract Borel, maximal-parabolic and boxed-kernel names are separate obligations; so are the final retained-theorem and source-contract synthesis. Root suggested using explicit coordinate and type actions and keeping kernel/splitting distinct from order agreement; this implementation and finite argument are the present author's work. The earlier construction, source-join and palette proofs and their distinct reviews are acquired inputs. No novelty, new census, current-status, Lean, measurement or whole-source credit is asserted.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* PSL(3, 4), Halle preprint 99–30, fixed 22-page edition, especially pp. 4 and 7–13.

[2] Magma handbook, *Permutation representations of linear groups*, ProjectiveSigmaLinearGroup/PSigmaL.
`https://magma.maths.usyd.edu.au/magma/handbook/text/713`.